

The Principle of Conservation of Mass:

$$\frac{D}{Dt} \int_{V_m} \rho(x,t) dV = 0$$

Raynolds

Transport Theorem:
$$\frac{D}{Dt} \int_{V_m} T(x,t) dV = \int_{V_c} \frac{\partial T(x,t)}{\partial t} dV + \int_{S_c} T(v \cdot n) dS$$

$$= \int_{V_c} \left(\frac{D T(x,t)}{Dt} + T \operatorname{div} v \right) dV$$

$$0 = \frac{D}{Dt} \int_{V_m} \rho(x,t) dV$$

$$0 = \int_{V_c} \frac{\partial \rho(x,t)}{\partial t} dV + \int_{S_c} \rho(v \cdot n) dS$$

$$\boxed{\int_{V_c} \frac{\partial \rho(x,t)}{\partial t} dV = - \int_{S_c} \rho(v \cdot n) dS}$$

(1) integral form of the conservation of mass in deformed config.

the rate of increase in mass = mass influx through the boundaries inside the control volume

Divergence Theorem
$$\int_{S_c} \rho(v \cdot n) dS = \int_{V_c} \operatorname{div}(\rho v) dV$$

$$(1): \int_{V_c} \left\{ \frac{\partial \rho}{\partial t} + \operatorname{div}(\rho v) \right\} dV = 0$$

$$\Rightarrow \boxed{\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho v) = 0}$$

Partial derivative of the conservation of mass in deformed config.

$$\frac{D(\cdot)}{Dt} = \frac{\partial(\cdot)}{\partial t} + \nabla(\cdot) \cdot \mathbf{r} \rightarrow \underbrace{\frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial x_i} v_i}_{\frac{D\rho}{Dt}} + \rho \frac{\partial v_i}{\partial x_i} = 0$$

$$\Rightarrow \boxed{\frac{D\rho}{Dt} + \rho \operatorname{div} \mathbf{v} = 0} \quad \text{Material derivative of the conservation of mass in deformed configuration}$$

Example 7.5.1

$$\begin{cases} x_1 = (1+\alpha t)X_1 \\ x_2 = X_2 \\ x_3 = X_3 \end{cases}, \quad \rho = \frac{\rho_0}{(1+\alpha t)}, \quad \rho_0 = \text{cte}$$

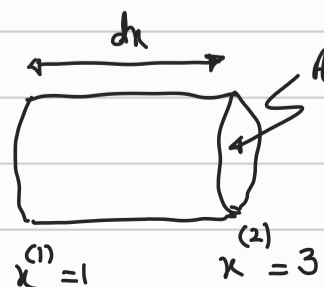
$$(a) \text{ velocity field, } v_i = \frac{Dx_i}{Dt} = \begin{cases} \frac{Dx_1}{Dt} = \alpha X_1 = \alpha \frac{x_1}{(1+\alpha t)} \\ \frac{Dx_2}{Dt} = 0 \\ \frac{Dx_3}{Dt} = 0 \end{cases}$$

(b) Check the continuity,

$$\begin{cases} \frac{D\rho}{Dt} + \rho \operatorname{div} \mathbf{v} = 0 \\ \frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{v}) = 0 \end{cases}$$

$$\begin{aligned} & \frac{\partial}{\partial t} \left(\frac{\rho_0}{1+\alpha t} \right) + \frac{\partial}{\partial x_i} \left(\frac{\rho_0}{1+\alpha t} \right) v_i + \left(\frac{\rho_0}{1+\alpha t} \right) \frac{\partial v_i}{\partial x_i} \\ &= \frac{-\rho_0 \alpha}{(1+\alpha t)^2} + \frac{\rho_0}{(1+\alpha t)} \frac{\alpha}{(1+\alpha t)} = 0 \quad \checkmark \end{aligned}$$

(c) total mass
rate of increase of mass



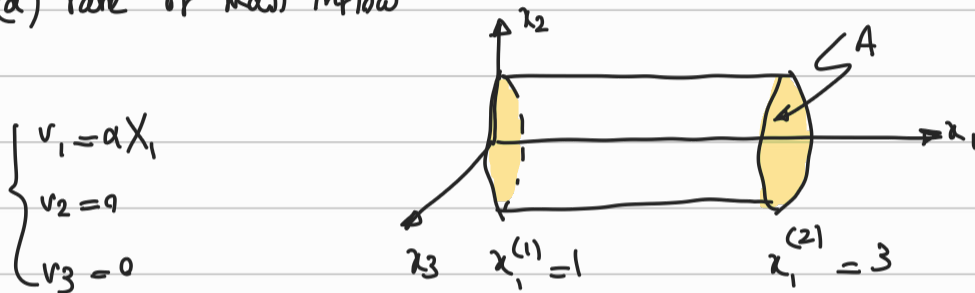
$$m(x_1, x_2, t) = \int_{x_1^{(1)}}^{x_1^{(2)}} \rho(x, t) dV = \int_1^3 \frac{\rho_0}{1+\alpha t} A dx_1 = \frac{\rho_0 A}{1+\alpha t} x_1 \Big|_1^3$$

$$= \frac{2 \rho_0 A}{(1+\alpha t)}$$

$$\frac{\partial m}{\partial t} = \frac{\partial}{\partial t} \left(\frac{2 \rho_0 A}{(1+\alpha t)} \right) = \frac{-2 \alpha \rho_0 A}{(1+\alpha t)^2}$$

mass influx (mass is increasing)

(d) rate of mass inflow



$$\text{mass inflow from } x_1^{(1)}: (\rho A v)_{x_1^{(1)}} = \frac{-\rho_0}{1+\alpha t} A \alpha x_1$$

$$= \frac{-\rho_0}{1+\alpha t} A \alpha \frac{x_1^{(1)}}{(1+\alpha t)} = \frac{-\rho_0 A \alpha}{(1+\alpha t)^2} x_1^{(1)} = \frac{-\rho_0 A \alpha}{(1+\alpha t)^2}$$

$$\text{mass outflow from } x_1^{(1)}: (\rho A v)_{x_1^{(2)}} = \frac{\rho_0}{1+\alpha t} A \alpha x_1$$

$$= \frac{3 \rho_0 A \alpha}{(1+\alpha t)^2}$$

$$\text{Net rate mass inflow} = \frac{3 \rho_0 A \alpha}{(1+\alpha t)^2} - \frac{\rho_0 A \alpha}{(1+\alpha t)^2} = \frac{2 \rho_0 A \alpha}{(1+\alpha t)^2}$$

(e) total mass at time t

reference time (t=0) was in control volume

$$\begin{aligned}
 M(X_1, X_2, t) &= \int_{x_1^{(1)}(X_1^{(1)}, t)}^{x_1^{(2)}(X_1^{(2)}, t)} \rho(x, t) dV = \int_{(1+\alpha t)X_1^{(1)}}^{(1+\alpha t)X_1^{(2)}} \frac{\rho_0}{(1+\alpha t)} A dx_1 \\
 &= \int_{1(1+\alpha t)}^{3(1+\alpha t)} \frac{\rho_0}{(1+\alpha t)} A dx_1 = \frac{\rho_0 A}{(1+\alpha t)} x_1 \Big|_{1(1+\alpha t)}^{3(1+\alpha t)} \\
 &= \frac{\rho_0 A}{(1+\alpha t)} 2(1+\alpha t) = 2\rho_0 A
 \end{aligned}$$

(f) total linear momentum of the fixed part of the material in part (e)

reference config. $\uparrow$

$$\int_{x_1^{(1)}}^{x_1^{(2)}} \rho(x, t) v_i dV = \int_{x_1^{(1)}}^{x_1^{(2)}} \frac{\rho_0}{(1+\alpha t)} v_i \hat{e}_i (A dx_1)$$

$$= \hat{e}_1 \int_{1(1+\alpha t)}^{3(1+\alpha t)} \frac{\rho_0 A}{(1+\alpha t)} \frac{\alpha x_1}{(1+\alpha t)} dx_1$$

$$= \hat{e}_1 \int_{(1+\alpha t)}^{3(1+\alpha t)} \frac{\rho_0 A \alpha}{(1+\alpha t)^2} x_1 dx_1$$

$$= \hat{e}_1 \frac{\rho_0 A \alpha}{(1+\alpha t)^2} \frac{x_1^2}{2} \Big|_{(1+\alpha t)}^{3(1+\alpha t)} = \hat{e}_1 \frac{\rho_0 A \alpha}{2(1+\alpha t)^2} \frac{4}{8(1+\alpha t)^2}$$

$$= (4\rho_0 A \alpha) \hat{e}_1$$